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1. Introduction

In today's educational environment, attendance management is important for ensuring student accountability and maintaining institutional records. However, many colleges still rely on manual methods such as calling names or signing attendance sheets, which are inefficient, time-consuming and often lead to human errors or cheating. To solve this issue, this project develops an IoT-Based Smart Attendance System called ATTENDLY that automatically records student attendance using IoT sensors and electronic identification. The data is uploaded to a cloud-based platform where lecturers can monitor records through a dashboard interface. By automating the process with IoT technology, the system reduces manual effort, minimizes errors, and improves the efficiency of attendance management in the college.

1.1. Problem Statement

Nowadays, most educational institutions still rely on manual attendance systems, which is inefficient, time consuming, and often lead to human error or cheating. These traditional systems lack real-time monitoring and timely updates, making it difficult for lecturers to track attendance effectively.

Therefore, there is a need for a low-cost, reliable, and IoT-based automated system that can ensure accurate and secure attendance logging with minimal human involvement while providing instant notifications to relevant personnel for real-time awareness and allowing lecturers to monitor real-time attendance reports. This system can also improve efficiency by reducing manual work and minimizing human errors.

1.2. Importance of the Problem

Addressing this problem is important because accurate attendance management can ensure operational efficiency and accountability within institutions. Inefficient or unreliable attendance systems can lead to data inconsistencies, unauthorized modifications and administrative challenges that compromise the credibility of institutional records.

By developing a low cost, automated, and IoT-based attendance solution, organizations can achieve greater accuracy and transparency in attendance reporting. Furthermore, automating this process minimizes human intervention and enhances overall productivity in both academic and professional environments.

1.3. Proposed Solution

Solution for Students :

The Smart Attendance System provides a fast, secure and user-friendly method for students to record their attendance using RFID technology. Each student is assigned a unique RFID card, which they scan upon entering the lecture hall. The system instantly verifies the student's identity with a pre-registered student database saved in Firebase and logs the attendance record automatically with a timestamp.

To improve reliability, the system is integrated with a motion sensor that activates the scanning process only when movement is detected, reducing unnecessary power usage. Additionally, visual feedback is provided through an LCD display and LED indicators to inform students whether their attendance has been successfully recorded.

Students will also receive a confirmation message when their attendance is successfully recorded, a reminder if they have not taken attendance within 10 minutes after the class begins, and an absence notification if attendance is still not recorded within 30 minutes after the start of the class. This real-time notification system helps students stay aware of their attendance status.

This solution minimizes common issues such as proxy attendance and human error, ensuring accurate data collection. It is also highly practical and realistic, as RFID technology is low-cost, widely available, and easy for students to use without requiring technical knowledge.

Solution for Lecturers :

For teachers, the system offers a centralized and automated attendance management platform. Attendance data collected via the ESP32 microcontroller is sent in real time to a cloud database (Firebase), where it is securely stored and organized.

The system also supports integration with Google Sheets for automated report generation, enabling teachers to track attendance rate for students and generate daily attendance charts.

This approach significantly reduces administrative workload, eliminates manual record-keeping errors, and enhances overall efficiency. The solution is highly justified as it leverages IoT technology to provide real-time, accurate and scalable attendance monitoring, which is essential for modern educational environments.

2. Objectives

This project focuses on **three key objectives** to ensure high quality and better efficiency.

Objective 1 (Identify):

To identify the limitations of traditional attendance systems such as inefficiency, inaccuracy, and time consumption so we can build a strong plan to fix it.

Objective 2 (Develop):

To develop a Smart Attendance System that can automatically record attendance using digital technology that improves the accuracy and efficiency of attendance management.

Objective 3 (Evaluate):

To evaluate the system performance based on accuracy, response time, reliability through system testing and user feedback.

3. Scope of the Project

3.1. Scope Of Implementation

The scope of implementation of this project is shown as below:

- Real-time attendance logging.
- Low-cost hardware setup.
- Remote dashboard access for lecturers.
- Real-time notification for students status

3.2. Scope Of Capabilities

The scope of capabilities of this project is shown as below:

- The system can automatically identify and record attendance using RFID cards without manual input.
- Attendance data is processed and updated instantly in the system's database, allowing lecturers to monitor attendance in real time.
- The system minimizes human errors and ensures accuracy in attendance records, preventing issues like duplication attendance.
- It can send automated notifications or alerts to students regarding attendance status.

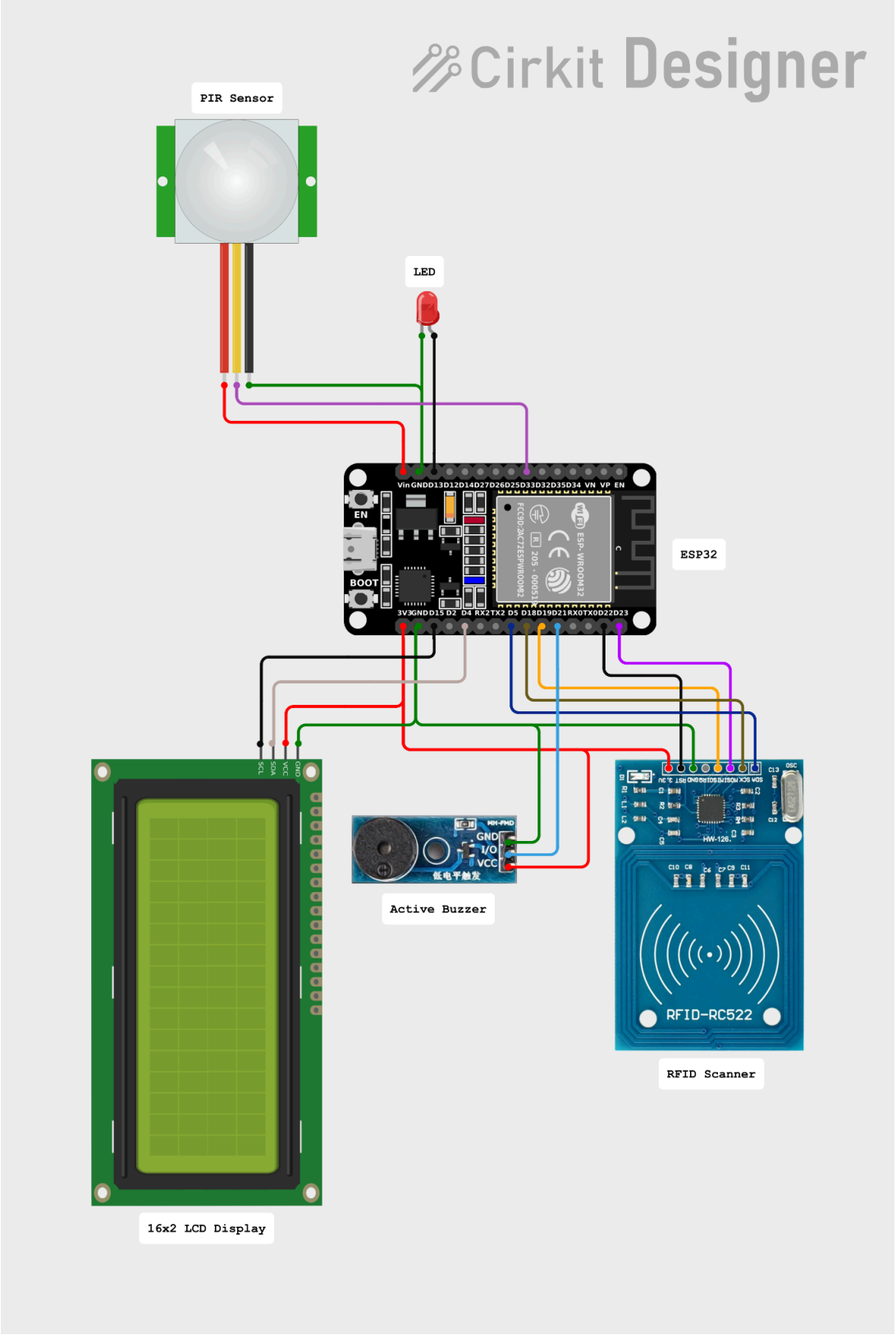
3.3. Limitation of Project

There are also several limitations that are present in this project:

- The system's performance heavily relies on hardware components such as RFID readers and PIR sensors. Any malfunction or poor maintenance can affect accuracy.
- The system is cloud-based, a stable internet connection is required for data synchronization and real-time updates. Offline operation may be limited.
- The implementation of smart technologies and hardware components may involve high initial costs for installation and configuration.

4. Methodology

4.1. Circuit Diagram



4.2. Flowchart

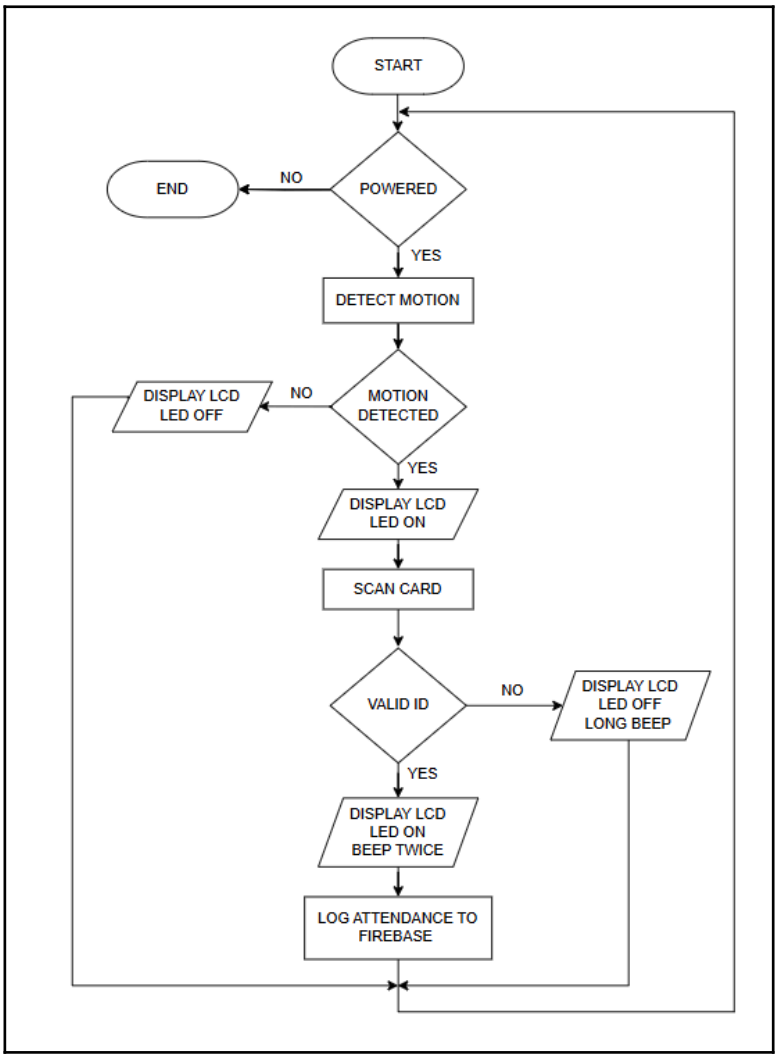


Figure 4.2.a

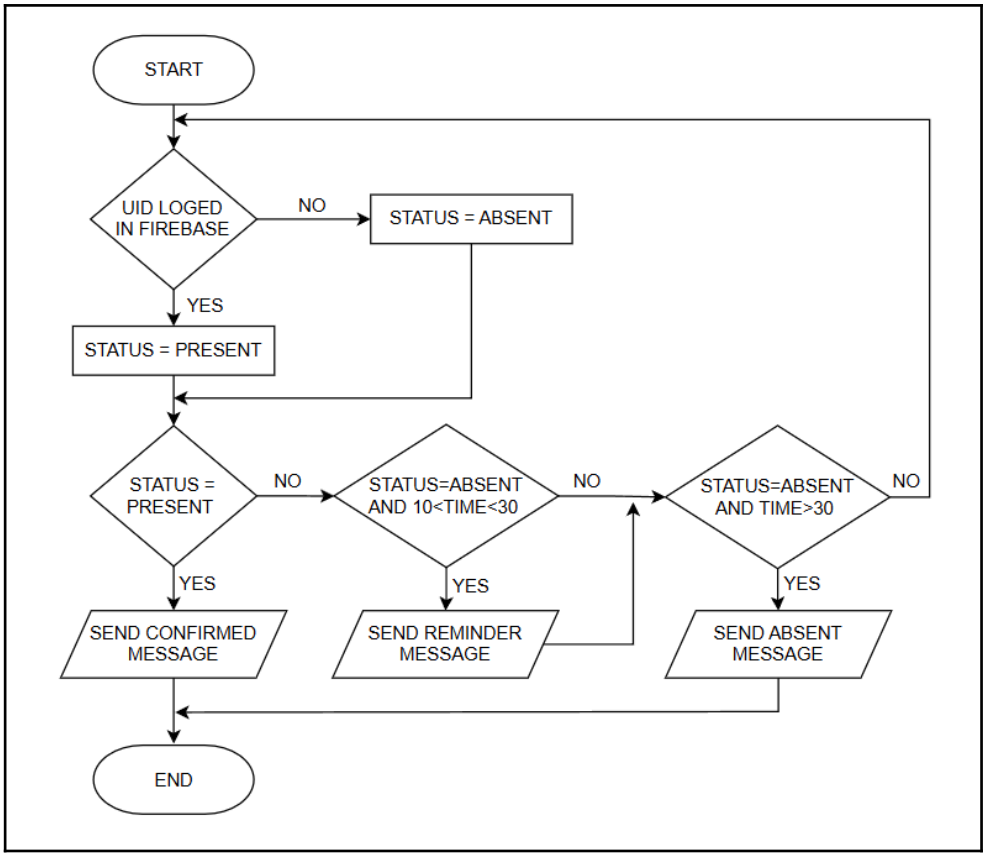
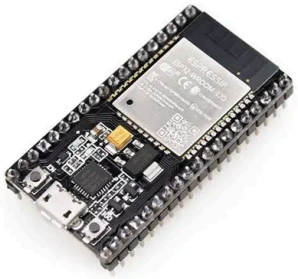
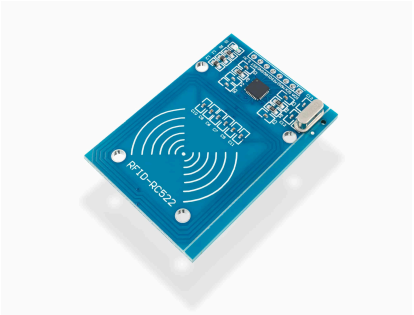
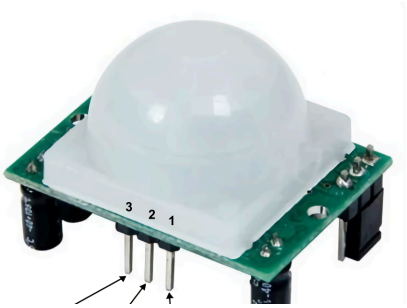
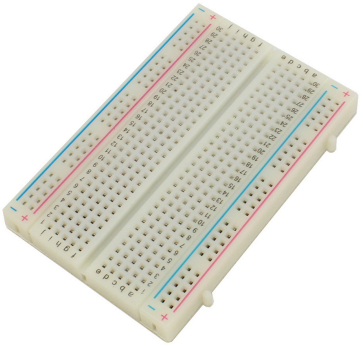
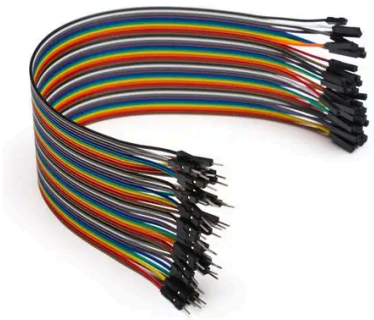

START
UID LOGED IN FIREBASE
NO
STATUS = ABSENT
YES
STATUS = PRESENT
STATUS = PRESENT
NO
STATUS=ABSENT AND 10<TIME<30
NO
STATUS=ABSENT AND TIME>30
NO
YES
YES
YES
SEND CONFIRMED MESSAGE
SEND REMINDER MESSAGE
SEND ABSENT MESSAGE
END

Figure 4.2.b

5. Specifications

5.1. Hardware

	**NodeMCU ESP32 (Wi-Fi built-in)** ● Perform the operation of the program
	**RFID Reader (MFRC522)** ● Read RFID card
	**PIR Sensor (HC-SR501)** ● Detect motion
	**Active Buzzer** ● Emits double beep sound if card ID scanned is valid ● Emits long beep sound if card ID scanned is invalid

	**16x2 Liquid-Crystal Display (LCD)** ● Display motion status ● Display student's name and success status if attendance is taken successfully ● Display error message if card ID scanned is invalid
	**LED** ● Emits light if there is motion ● Emits light if card ID scanned is valid
	**Breadboard** ● Used for component connections
	**Jumper Wire** ● Connects components on the breadboard

	**Power Bank** ● Act as a power supply to all electronic components
	**RFID Card (MIFARE)** ● Used by students to take attendance

5.2. Software

	**Arduino IDE** • Used to program the ESP32 microcontroller. • Store attendance data to firebase. • Control the logic of the system.
	**Firebase** • Used as a real-time cloud database to store and manage attendance data.
	**Google Sheets** • Organize and display attendance data in table form. • Helps lecturers easily view, track, and analyze student attendance records with charts.
	**Google Apps Script** • Connect Firebase with Google Sheets to automatically transfer and update attendance records. • Trigger Telegram Bot notifications (confirmation, reminders, absence alerts) automatically based on attendance data.

	**Telegram BotFather** • Create a telegram bot to send automatic notifications to students. • Sends messages such as attendance confirmation, reminders, and absence alerts in real time.
	**Android Studio** • Develop the mobile application (ATTENDLY), which provides an alternative platform for lecturers to view and manage attendance records, instead of relying solely on Google Sheets.

6. Implementation

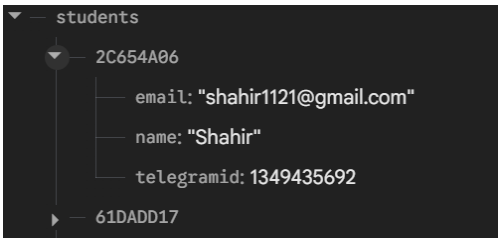
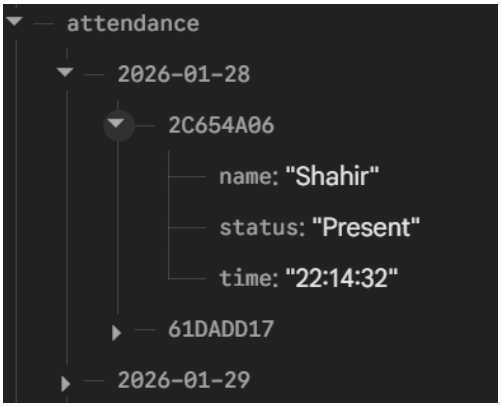
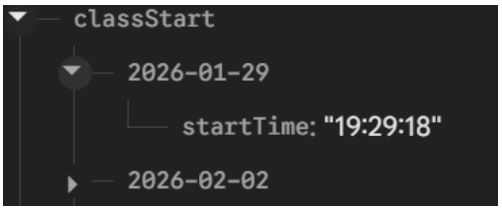
6.1. Implementation of The Design Solution

The Smart Attendance System (Attendly) was implemented with three main components:

Firebase Database	Stores student and attendance data
Arduino (ESP32)	Handles hardware interaction, RFID scanning and data recording to firebase
Google Apps Script	Automates attendance processing, ensuring accurate synchronization between Firebase and Google Sheets, and providing real time notifications.

6.1.1. Firebase Realtime Database Structure

Firebase acts as the central data storage system designed based on the system requirements. The database is structured into three main nodes:

	**/students/** ● Stores registered students ● Used for validation to check if UID exists
	**/attendance/** ● Record attendance for valid students ● Stores student's name, timestamp, and attendance status
	**/classStart/** ● Logged when the system activated ● Stores session start time ● Used to identify absent status of students for reminder message

6.1.2. Arduino Implementation

The Arduino implementation using the ESP32 acts as the main controller of the Smart Attendance System. During operation, the system continuously runs inside the loop() function. First, the PIR sensor detects motion. If motion is detected, the system activates, turns on the LED indicator and prompts the user to scan their RFID card using the LCD display. If no motion is detected, the system remains idle and skips further processing.

When a card is scanned, the system reads the RFID UID and converts it into a readable format.The UID is then checked against the Firebase database under the /students/ node to verify whether the UID is registered.

If the UID exists in the database, the system retrieves the associated student name, displays a confirmation message on LCD along with the student's name. A double beep sound is also emitted to indicate successful attendance recording. Subsequently, the system executes the attendance recording function, where it generates the current date and time using the synchronized NTP clock and stores the data in Firebase under the /attendance/ node. The stored data includes the student's name, timestamp, and attendance status.

Conversely, if the UID is not found in the database or if a retrieval error occurs, the LCD displays an "Access Denied" message, and a long buzzer sound is triggered to alert the user of an invalid card. This ensures that only registered users are allowed to record attendance, maintaining system integrity.

After each process, the system resets and continues scanning for the next user. Overall, the Arduino implementation follows the same sequence and logic as the flowchart in Figure 4.2.a, ensuring that the system design is correctly implemented.

6.1.3 Google Apps Script Implementation

The Google Apps Script is implemented as the backend processing component of the Smart Attendance System, responsible for retrieving, organizing, and managing attendance data from Firebase to Google Sheets. A function called runDailyAttendanceSystem() first obtains the current date and fetches the class start time from the /classStart/ node in Firebase. If no start time is found, the script stops execution, ensuring that attendance processing only begins after the ESP32 has initialized the session.

The script then processes each student by comparing their UID saved in Google Sheets with the Firebase attendance data. If a match is found, the student is marked as “Present” and their scan time is recorded; otherwise, they are marked as “Absent.” The results are then saved in the “Attendance” sheet. If a record for the student does not exist for the current date, a new row is added; if it already exists, the record is updated. This ensures that the data remains accurate and prevents duplication.

In addition, the script implements an automated notification system. If a student has successfully scanned their card, a confirmation Telegram message is sent. If a student has not scanned within 10 minutes after the class start time, a reminder notification is sent. If the student is still absent after 30 minutes, a final absence notification is also sent. This logic follows the decision-making structure in the algorithm, where different actions are triggered based on time and attendance status.

Furthermore, a function called uploadToFirebase() allows student data from Google Sheets to be uploaded to Firebase under the /students/ node, ensuring that the database remains updated with registered users.

Overall, the Google Apps Script implementation fully supports the system design by automating attendance processing, ensuring accurate synchronization between Firebase and Google Sheets, and providing timely notifications.

6.2. Testing approach/Strategy and Design

The testing strategy for the Smart Attendance System was designed to ensure that all system components, including the ESP32 hardware, Firebase database, and Google Apps Script to function correctly and reliably. A structured approach was used by dividing the testing process into three main levels: unit testing, integration testing, and system testing.

Firstly, unit testing was conducted on individual components to verify their functionality. For the Arduino (ESP32), each hardware module such as the RFID reader, PIR sensor, LCD display, buzzer, and WiFi connection was tested separately. For example, the RFID module was tested to ensure it could correctly read UID values, while the PIR sensor was tested to confirm accurate motion detection. Similarly, Firebase connectivity was tested to ensure data could be successfully sent and retrieved. For Google Apps Script, individual functions such as data retrieval, attendance processing, and notification sending were tested independently to ensure correct logic execution.

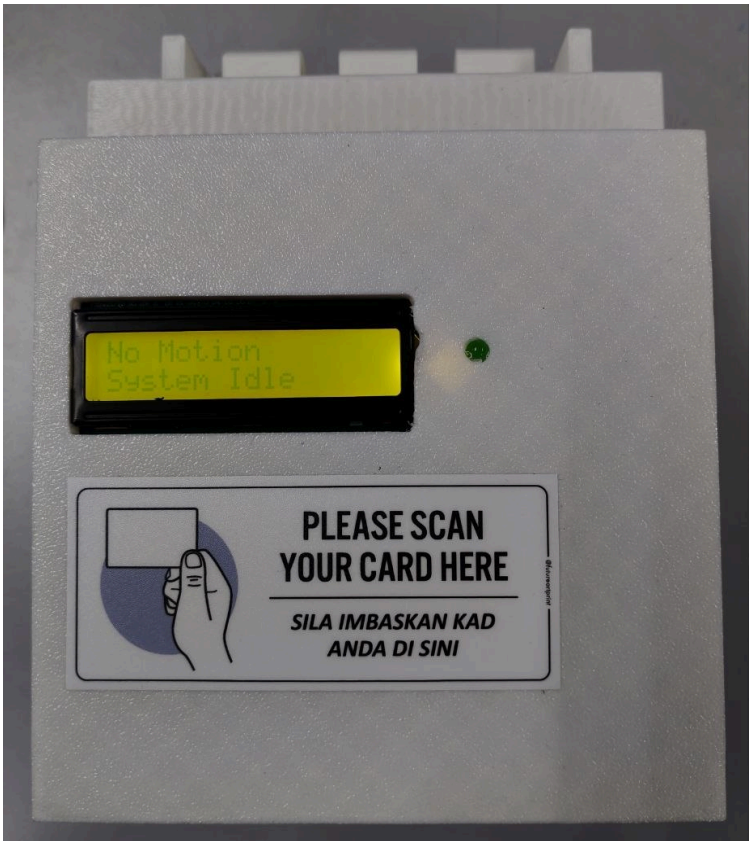
Secondly, integration testing was performed to verify that different components of the system work together as expected. This includes testing the communication between ESP32 and Firebase, where scanned RFID data is stored in real-time, as well as the interaction between Firebase and Google Apps Script, where attendance data is retrieved and processed into Google Sheets. This step ensures that data flows correctly across all system components without errors, data loss or duplicate data.

Finally, system testing was carried out to evaluate the overall performance of the complete system in a real-world scenario. This involves testing the full process from motion detection, RFID scanning, data verification, attendance recording, and notification delivery. Different test scenarios were designed, such as valid card scanning, invalid card detection, late attendance and absence cases. The results were then compared with the expected outcomes to determine whether the system meets its functional requirements.

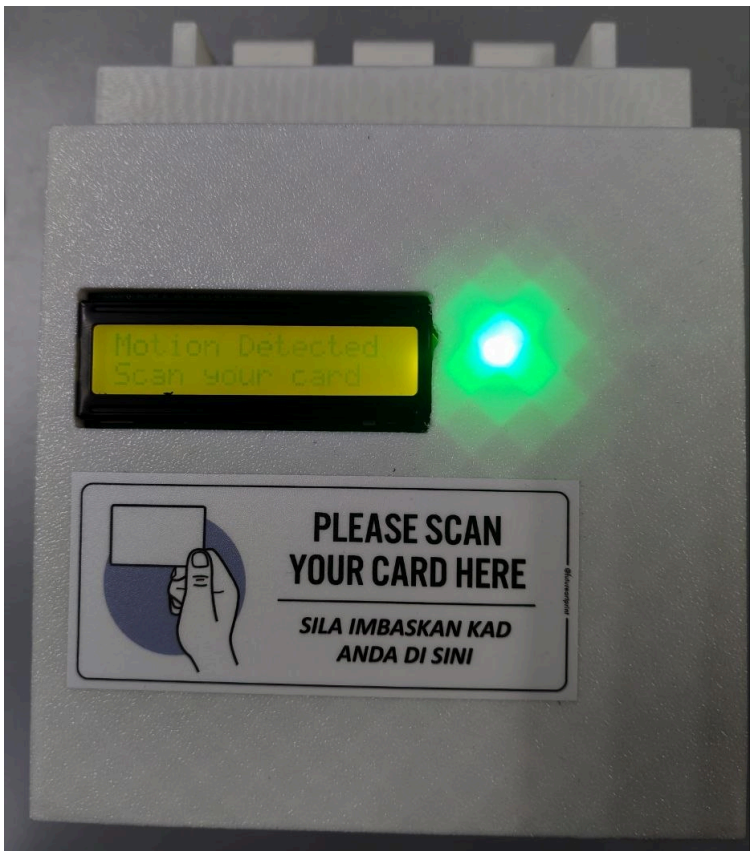
Overall, the testing strategy is comprehensive and well-structured, covering all levels of the system and ensuring that the implementation is reliable, accurate, and aligned with the system design.

6.3. Result & Discussion

The results of the Smart Attendance System demonstrate that the proposed solution functions effectively in automating attendance recording and monitoring. The system was tested under multiple scenarios, including valid card scanning, invalid card detection, late attendance, and absence handling. The outcomes were recorded and compared with the expected results to evaluate system performance.



When no motion detected



When motion is detected



Invalid Card Detection



Valid card scanned

ATTENDLY Dashboard

File Edit View Insert Format Data Tools Extensions Help

Share

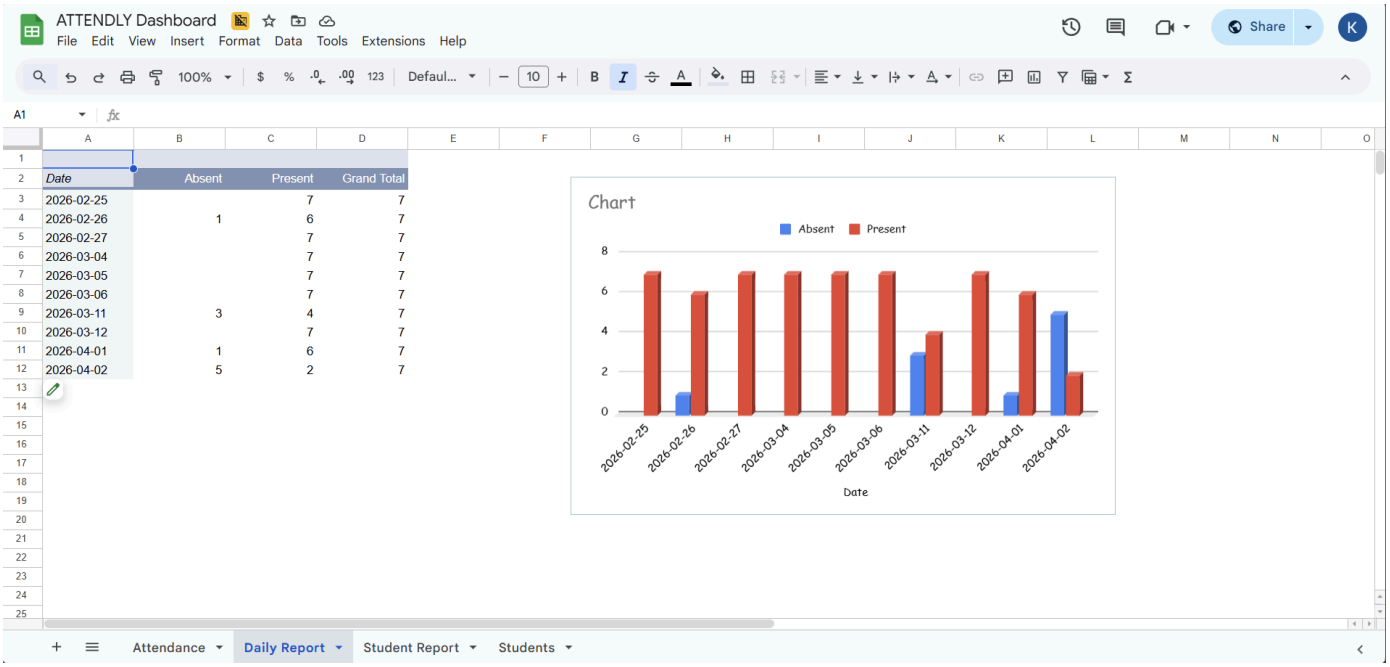
Attendance

Name	UID	Date	Time	Status	Mail Control
Afiq	61DADD17	2026-02-25	9:19:26	Present	SCANNED_MAIL_SENT
Irfan	7219404C	2026-02-25	9:19:09	Present	SCANNED_MAIL_SENT
Khor	228F484C	2026-02-25	9:19:21	Present	SCANNED_MAIL_SENT
Shahir	A2FE434C	2026-02-25	9:18:39	Present	SCANNED_MAIL_SENT
Jordan	D223194C	2026-02-25	9:19:44	Present	SCANNED_MAIL_SENT
Ali	2206164C	2026-02-25	9:21:05	Present	SCANNED_MAIL_SENT
Abu	520CCC4B	2026-02-25	9:19:51	Present	SCANNED_MAIL_SENT
Afiq	61DADD17	2026-02-26	10:27:38	Present	SCANNED_MAIL_SENT
Irfan	7219404C	2026-02-26	10:28:16	Present	SCANNED_MAIL_SENT
Khor	228F484C	2026-02-26	10:28:32	Present	SCANNED_MAIL_SENT
Shahir	A2FE434C	2026-02-26	10:27:28	Present	SCANNED_MAIL_SENT
Jordan	D223194C	2026-02-26	10:28:39	Present	SCANNED_MAIL_SENT
Ali	2206164C	2026-02-26	--	Absent	ABSENT_MAIL_SENT
Abu	520CCC4B	2026-02-26	10:28:44	Present	SCANNED_MAIL_SENT
Afiq	61DADD17	2026-02-27	8:22:55	Present	SCANNED_MAIL_SENT
Irfan	7219404C	2026-02-27	8:23:42	Present	SCANNED_MAIL_SENT
Khor	228F484C	2026-02-27	8:23:37	Present	SCANNED_MAIL_SENT
Shahir	A2FE434C	2026-02-27	8:22:46	Present	SCANNED_MAIL_SENT
Jordan	D223194C	2026-02-27	8:23:31	Present	SCANNED_MAIL_SENT
Ali	2206164C	2026-02-27	8:23:23	Present	SCANNED_MAIL_SENT
Abu	520CCC4B	2026-02-27	8:23:11	Present	SCANNED_MAIL_SENT

ATTENDLY
Smart Presence, Simplified.

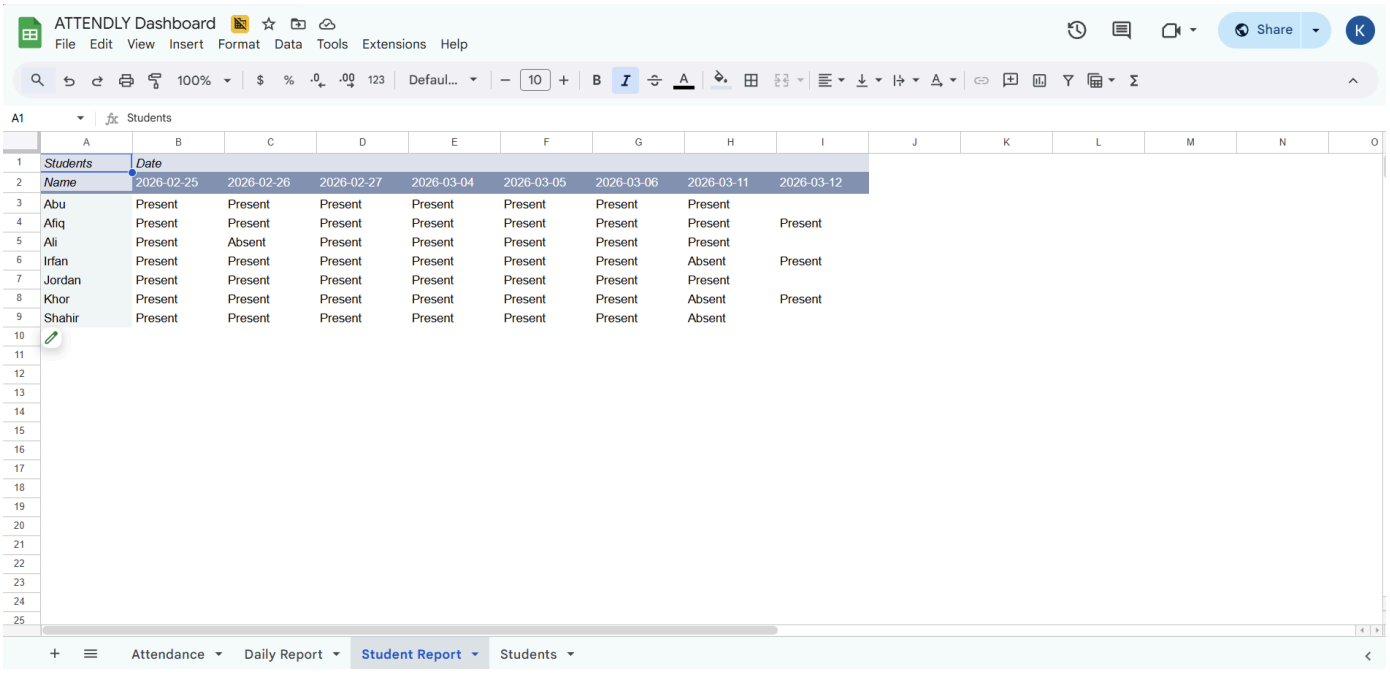
Attendance | Daily Report | Student Report | Students

Attendance Sheet



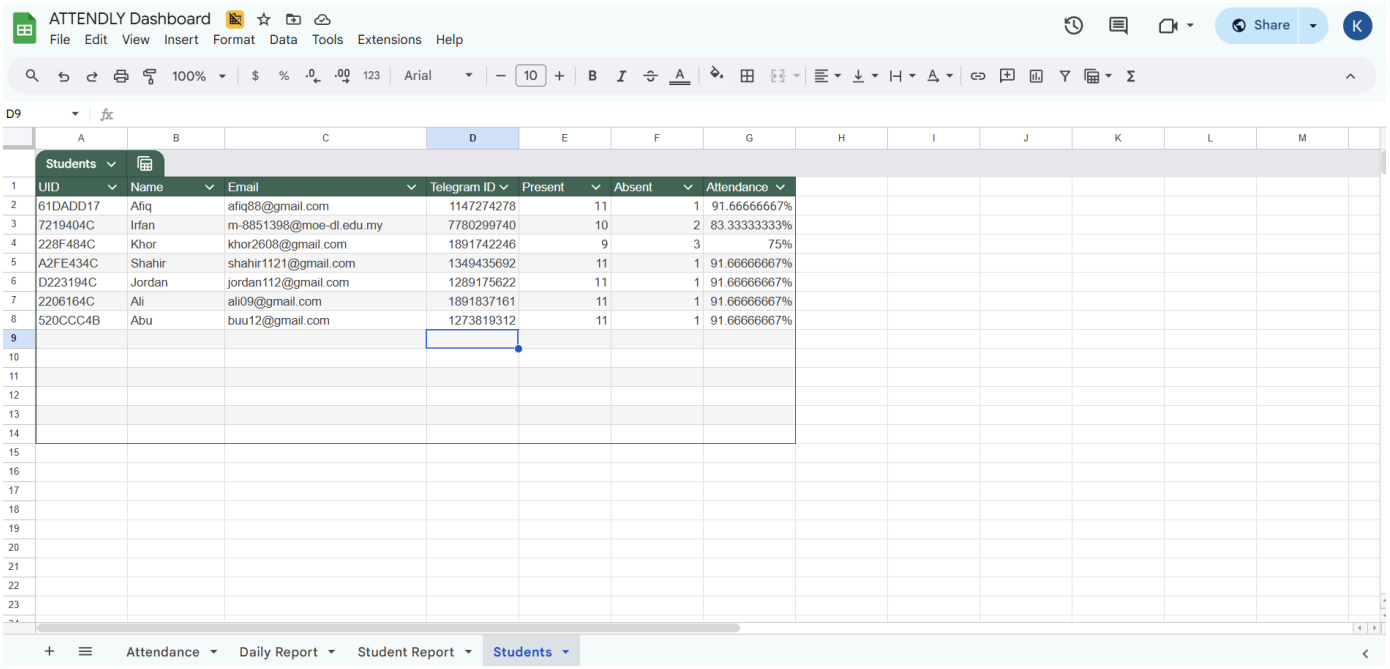
Date	Absent	Present	Grand Total
2026-02-25		7	7
2026-02-26	1	6	7
2026-02-27		7	7
2026-03-04		7	7
2026-03-05		7	7
2026-03-06		7	7
2026-03-11	3	4	7
2026-03-12		7	7
2026-04-01	1	6	7
2026-04-02	5	2	7

Daily Attendance Insights



Students	Date							
Name	2026-02-25	2026-02-26	2026-02-27	2026-03-04	2026-03-05	2026-03-06	2026-03-11	2026-03-12
Abu	Present	Present	Present	Present	Present	Present	Present	
Afiq	Present	Present	Present	Present	Present	Present	Present	Present
Ali	Present	Absent	Present	Present	Present	Present	Present	
Irfan	Present	Present	Present	Present	Present	Present	Absent	Present
Jordan	Present	Present	Present	Present	Present	Present	Present	
Khor	Present	Present	Present	Present	Present	Present	Absent	Present
Shahir	Present	Present	Present	Present	Present	Present	Absent	

Student Attendance Record


ATTENDLY Dashboard

File Edit View Insert Format Data Tools Extensions Help

Share

Students

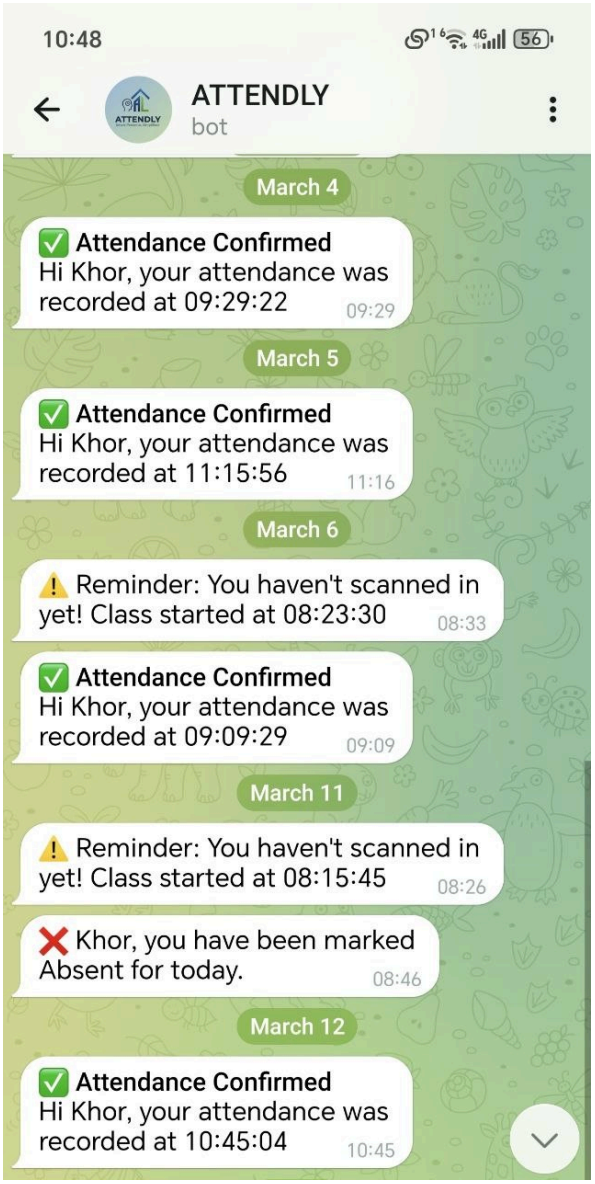
UID	Name	Email	Telegram ID	Present	Absent	Attendance
61DADD17	Afiq	afiq88@gmail.com	1147274278	11	1	91.66666667%
7219404C	Irfan	m-8851398@moe-dl.edu.my	7780299740	10	2	83.33333333%
228F484C	Khor	khor2608@gmail.com	1891742246	9	3	75%
A2FE434C	Shahir	shahir1121@gmail.com	1349435692	11	1	91.66666667%
D223194C	Jordan	jordan112@gmail.com	1289175622	11	1	91.66666667%
2206164C	Ali	ali09@gmail.com	1891837161	11	1	91.66666667%
520CCC4B	Abu	buu12@gmail.com	1273819312	11	1	91.66666667%

Attendance Daily Report Student Report Students

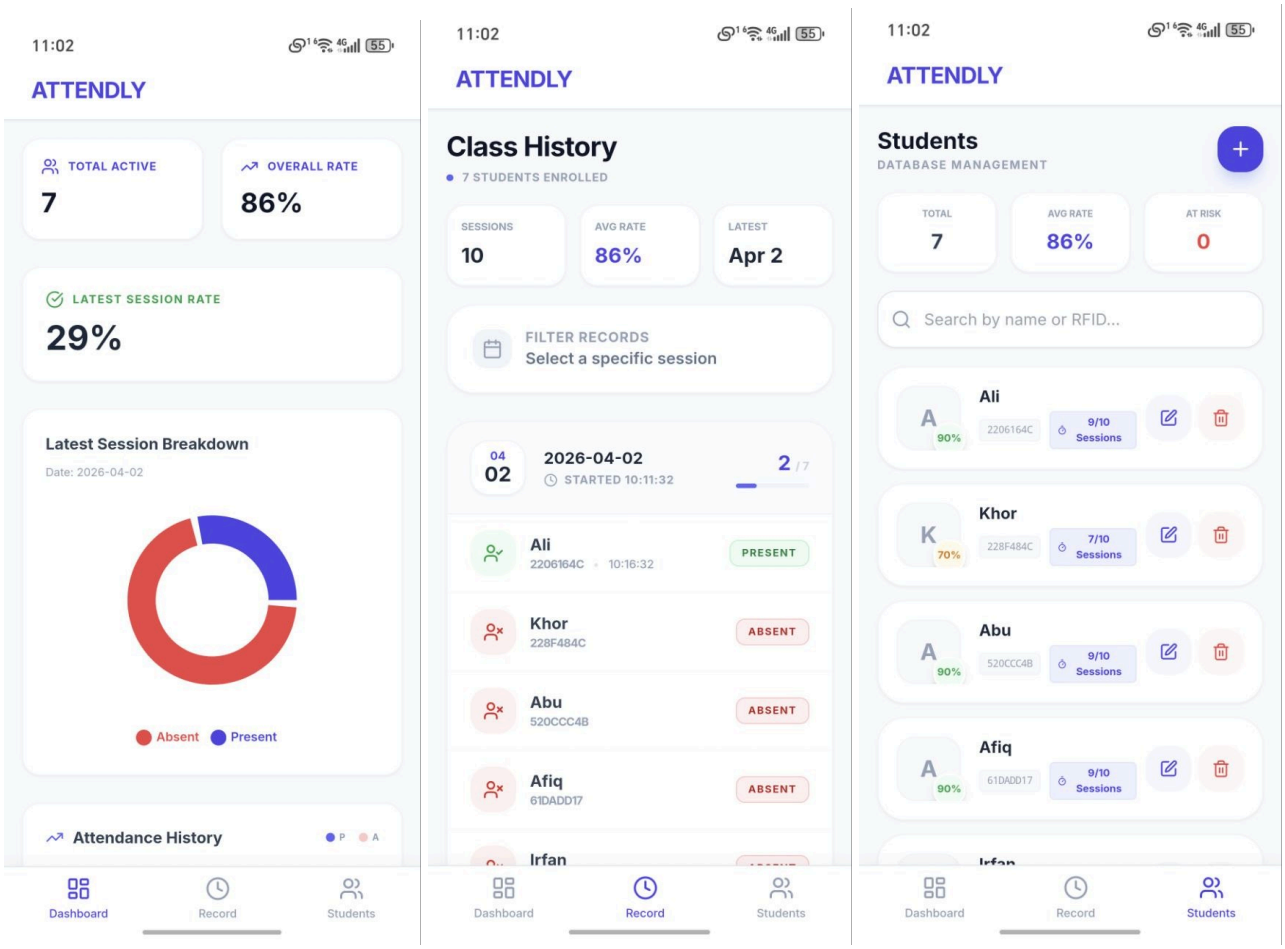
Student Sheet



Attendly Dashboard Google Sheets



Telegram Reminder Notification



Attendly Application

Besides, the Attendly application(Android) is implemented to provide an alternative dashboard for the lecturer to access attendance records. The application retrieves data directly from the Firebase Realtime Database, ensuring that all attendance information displayed is updated in real time. This allows lecturers to monitor attendance without relying solely on Google Sheets.

Scan to download

The system achieved a high level of accuracy in detecting and recording attendance. RFID scanning was reliable, with UID values correctly read and matched against the database. The integration with Firebase allowed real-time data storage, while Google Apps Script ensured that the data was properly processed and displayed in a structured format.

From a critical perspective, the system shows strong performance in terms of automation, accuracy, and usability. The use of motion detection improves efficiency by activating the system only when needed, while NTP time synchronization ensures accurate timestamp recording. In addition, the automated notification feature enhances user engagement by informing students of their attendance status.

Despite these strengths, several limitations were identified. Firstly, the system relies heavily on internet connectivity; any disruption in WiFi may delay or prevent data transmission. Secondly, the RFID system requires physical cards, which may be lost or misused by users.

Based on the results, the system can be further improved by implementing offline data buffering to handle network failures. Overall, the results confirm that the Smart Attendance System meets its objectives and operates in accordance with the proposed design. The system is capable of accurately recording attendance, managing data efficiently, and providing meaningful feedback to users. The discussion highlights both the strengths and areas for improvement, demonstrating a comprehensive and critical evaluation of the system.

7. Conclusion

In conclusion, the ATTENDLY IoT-Based Smart Attendance System provides an efficient and reliable solution to the challenges of traditional attendance management. By integrating PIR and RFID sensors with a cloud-based platform, the system automates attendance recording, reduces human errors, and allows real-time monitoring for lecturers. With added features such as instant Telegram notifications, the system ensures students are informed of their attendance status, enhancing accountability and transparency. Overall, ATTENDLY simplifies the attendance process, improves accuracy, and increases productivity within educational institutions.

7.1. Future Work

Although the ATTENDLY system successfully improves attendance management, several enhancements can be implemented to further increase its effectiveness and scalability. One potential improvement is the integration of biometric authentication, such as fingerprint or facial recognition, to enhance security and prevent proxy attendance, which RFID systems alone may not fully eliminate.

In addition, the system can be upgraded by incorporating a mobile push notification feature within the ATTENDLY application itself, reducing dependence on third-party platforms like Telegram and providing a more centralized user experience.

Furthermore, the system can be expanded to support multi-classroom environments by improving its database structure and cloud scalability. Offline functionality could also be introduced to ensure the system continues to operate during internet disruptions, with automatic data synchronization once connectivity is restored.

Finally, future development may include enhancing hardware efficiency by optimizing sensor accuracy, making the system more cost-effective and suitable for large-scale deployment. These improvements would not only strengthen the reliability of the system but also increase its practicality in real-world educational environments.

8. References

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